CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (EE) Examination

May 2013

EE404 Energy Management and Conservation

Date: 09.05.2013, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain Classification of energy based on different criteria. [07]

Q - 2 (a) What is Energy? Explain energy science & energy Technology. [07]

(b) Draw Flow chart of energy planning. [03]

(c) Explain the law of energy conversion with the generalized equation. [04]

OR

Q - 2 Write Short notes: (Any Two) [14]

(1) System Power factor and use of capacitor.

(2) Electrical distribution system.

(3) Energy management.

Q - 3 (a) Draw Classification chart of furnaces and write characteristic of an efficient Furnace. [07]

(b) Explain Need for Electrical Load Management and Step By Step Approach for Maximum Demand Control. [07]

OR

Q - 3 (a) Explain energy conservation in any manufacturing industry. [07]

(b) Explain Responsibilities and duties of energy manager as per energy Conservation Act,2001. [07]

SECTION – II

Q - 4	(a)	Explain Topping cycles, Bottoming cycles and Combined cycles.	[07]
Q - 5	(a)	Write the Potential Energy Conservation Opportunities in Motor and Transformer.	[07]
	(b)	Suggest six points for conservation of energy.	[03]
	(c)	What is carbon credit? Explain Application of carbon credit.	[04]

OR

Q - 5	(a)	Write the Potential Energy Conservation Opportunities in HVAC System.	[07]
	(b)	Explain energy costing, energy organization and energy budgeting.	[03]
	(c)	Write Weather impact and pollution control impact on energy consumption.	[04]
Q - 6	(a)	Write the list of electrical system optimization and explain any two in details.	[07]
	(b)	Explain energy Audit of Illumination system.	[04]
	(c)	Explain energy Audit instruments.	[03]

OR

Q - 6	(a)	Explain detailed energy Audit.	[07]
	(b)	Explain Advance Flywheel Energy Storage.	[07]

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